

Test Plan — IFL Chapter Malang Website

Project: Website IFL Chapter Malang (Indonesian Future Leaders) **Repo:**

https://github.com/irzadzulhika29/fe_iflchaptermalang **Document Version:** v1.0 **Date:** 2026-06-22

Prepared By: Muhammad Irza Dzulhika (QA Intern Candidate) **Template:** Based on BINUS University Test Plan Framework (Mulyanto, 2018)

Document Information

Field	Value
Project Name	IFL Chapter Malang Website (website-ifl)
Document Version	v1.0
Date	2026-06-22
Prepared By	Muhammad Irza Dzulhika
Reviewed By	(to be assigned)
Approved By	(to be assigned)
Repo	github.com/irzadzulhika29/fe_iflchaptermalang
Live URL	https://iflchaptermalang.org/
Tech Stack	Vite 5 + React 18 + JavaScript

1. Introduction (Pendahuluan)

1.1. Rencana Pengujian

Dokumen ini menjabarkan rencana pengujian untuk **Website IFL Chapter Malang** — platform resmi chapter Indonesian Future Leaders di Malang. Website ini berfungsi sebagai pusat informasi chapter, platform donasi (dengan payment gateway Midtrans), portal blog/artikel, manajemen event, dan recruitment volunteer. Pengujian akan mencakup 7 modul fitur utama (authentication, blog, campaign, donation, event, profile, volunteer) + chatbot donasi sebagai fitur paling kompleks. **Status awal: 0% test coverage** — strategi pengujian akan dimulai dengan setup fondasi test (Vitest + Playwright + RTL) sebelum eksekusi.

1.2. Tujuan Pengujian

- **Membangun fondasi testing** dari 0% (saat ini tidak ada test file sama sekali)
- **Memvalidasi alur donasi end-to-end** (paling kritis karena ada payment gateway Midtrans)
- **Memverifikasi role-based access** (admin / bismar / copywriter) di 3 level proteksi

- **Mengurangi risiko regression** saat ada update konten / event baru
- **Memenuhi standar kualitas** untuk apply magang QA di organisasi IT/NGO

1.3. Metodologi Pengujian

Metode	Tools	Coverage Target
Black-box functional	Manual + Postman	Semua fitur user-facing
Component unit	Vitest + React Testing Library	Target 50% coverage dalam 3 bulan
End-to-end	Playwright	Critical paths: Auth, Donasi, Blog, Event
Security	OWASP ZAP, manual probing	Auth flow, role bypass, JWT
Payment testing	Midtrans sandbox	Transaction success/failure/timeout
A11y	axe-core (planned)	Form labels, keyboard nav
Visual regression	Percy / Chromatic (planned)	Public pages, donation flow

1.4. Objek Pengujian

Layer	Technology	Test Scope
Frontend	Vite 5.4 + React 18.2 (SPA)	Full UI + client logic
Routing	React Router DOM 6.17 (client-side)	33 routes (16 public, 5 auth, 10 protected, 2 error)
State	TanStack React Query 5.8	Server state caching
HTTP Client	Axios 1.7 + jwt-decode	All API calls
Auth	JWT (custom) + Google OAuth	Login, register, role guard
Rich text	CKEditor 5	Blog authoring
Image	ImageKit CDN	Image upload/transform
Payment	Midtrans Snap	Donation flow
Backend (out of scope)	External API (IFL backend)	E2E only

1.5. Arsitektur Sistem yang Diuji



1.6. Pembagian Sistem untuk Pengujian Unit dan Integrasi

Test Level	Target	Tools	Lokasi
Unit test	Pure functions, isolated components	Vitest + RTL	<code>src/**/__tests__/</code> (to be created)
Component integration	Component + React Query	Vitest + RTL	<code>src/**/__tests__/</code>
Hook test	Custom hooks (auth, blog, dll)	Vitest + RTL	<code>src/features/*/hooks/__tests__/</code>
API integration	Axios calls mocked via MSW	Vitest	<code>src/libs/api/__tests__/</code>
E2E	Full user flow	Playwright	<code>e2e/*.spec.ts</code> (to be created)
Manual	Visual, exploratory	Manual + screenshots	Staging

2. Test Items (Item Pengujian)

2.1. Item yang Akan Diuji

A. Authentication & Authorization (HIGH PRIORITY)

ID	Item	Tipe	Modul Path
AUTH-01	Login form (email + password)	UI/Functional	<code>src/features/authentication/</code>
AUTH-02	Register form (validation)	UI/Functional	<code>src/page/auth/RegisterPage.jsx</code>
AUTH-03	Forgot password flow	UI/Functional	<code>src/page/auth/ForgotPasswordPage.jsx</code>
AUTH-04	Email verification	Integration	<code>src/page/auth/VerifyPage.jsx</code>
AUTH-05	Google OAuth callback	Integration	<code>src/page/auth/GoogleCallbackPage.jsx</code>
AUTH-06	JWT token storage & refresh	Integration	<code>src/libs/api + jwt-decode</code>
AUTH-07	Logout (clear token, redirect)	UI/Functional	<code>src/features/authentication/</code>
AUTH-08	ProtectedToken route guard	Integration	<code>src/routes/ProtectedToken.jsx</code>
AUTH-09	ProtectedDashboard route guard	Integration	<code>src/routes/ProtectedDashboard.jsx</code>
AUTH-10	ProtectedRoles (admin/bismar/copywriter)	Security	<code>src/routes/ProtectedRoles.jsx</code>

B. Public Pages (MEDIUM PRIORITY)

ID	Item	Tipe	Modul Path
PUB-01	Home page (hero, about, programs)	UI/Visual	<code>src/page/root/HomePage.jsx</code>
PUB-02	About page	UI/Visual	<code>src/page/root/AboutPage.jsx</code>
PUB-03	Event listing	UI/Functional	<code>src/page/root/EventPage.jsx</code>
PUB-04	Project listing	UI/Functional	<code>src/page/root/ProjectPage.jsx</code>
PUB-05	Department pages (nondept, cdsi, bismar, dpp, hrd, comper)	UI/Visual	<code>src/page/root/</code>
PUB-06	Blog listing	UI/Functional	<code>src/page/root/BlogPage.jsx</code>
PUB-07	Single blog post (rich text)	UI/Functional	<code>src/page/root/SingleBlogPage.jsx</code>
PUB-08	Donation listing	UI/Functional	<code>src/page/root/DonationPage.jsx</code>
PUB-09	Single donation campaign	UI/Functional	<code>src/page/root/SingleDonationPage.jsx</code>
PUB-10	404 Not Found page	UI/Visual	<code>src/page/NotFoundPage.jsx</code>
PUB-11	Coming Soon page	UI/Visual	<code>src/page/ComingSoon.jsx</code>

C. Donation & Payment (HIGHEST PRIORITY — CRITICAL)

ID	Item	Type	Modul Path
DON-01	Browse donation campaigns	UI/Functional	src/features/donation/
DON-02	View campaign detail	UI/Functional	src/page/root/SingleDonationPage.jsx
DON-03	Start chatbot donation flow	Critical	src/page/root/ChatbotPage.jsx
DON-04	Chatbot composer (input)	UI/Functional	src/components/chatbot/chatcomposer/
DON-05	Chatbot quick replies	UI/Functional	src/components/chatbot/quickreply/
DON-06	Chatbot validation feedback	UI/Functional	src/components/chatbot/validationfeedback/
DON-07	Chatbot summary card	UI/Functional	src/components/chatbot/summarycard/
DON-08	Midtrans payment integration	Critical	src/libs/midtrans/
DON-09	Payment info display	UI/Functional	src/components/chatbot/paymentinfo/
DON-10	Payment success page	UI/Functional	src/page/root/PaymentSuccessPage.jsx
DON-11	Payment timeout/failure	Edge case	src/components/chatbot/successmodal/
DON-12	QRIS donation flow	UI/Functional	src/components/donationpayment/
DON-13	Donation confirmation modal	UI/Functional	src/components/chatbot/successmodal/

D. Blog & Content (MEDIUM PRIORITY)

ID	Item	Type	Modul Path
BLG-01	Blog listing (pagination)	UI/Functional	<code>src/features/blog/</code>
BLG-02	Single blog post view	UI/Functional	<code>src/page/root/SingleBlogPage.jsx</code>
BLG-03	Search blog posts	UI/Functional	<code>src/components/searchbar/</code>
BLG-04	Blog categories filter	UI/Functional	<code>src/features/blog/</code>
BLG-05	Admin: Add blog (CKEditor)	UI/Functional	<code>src/page/dashboard/AddBlogPage.jsx</code>
BLG-06	Admin: Edit blog	UI/Functional	<code>src/page/dashboard/EditBlogPage.jsx</code>
BLG-07	Admin: Delete blog	Integration	<code>src/features/blog/services/</code>
BLG-08	CKEditor 5 content authoring	UI/Functional	<code>@ckeditor/ckeditor5-react</code>
BLG-09	Blog image upload (ImageKit)	Integration	<code>imagekitio-react</code>
BLG-10	Blog categories management	UI/Functional	<code>src/page/dashboard/BlogCategories.jsx</code>
BLG-11	Blog detail admin view	UI/Functional	<code>src/page/dashboard/DetailBlogPage.jsx</code>

E. Event Management (MEDIUM PRIORITY)

ID	Item	Tipe	Modul Path
EVT-01	Public event listing	UI/Functional	src/page/root/EventPage.jsx
EVT-02	Event detail page	UI/Functional	src/features/event/
EVT-03	Admin: Event management	UI/Functional	src/page/dashboard/EventManagerPage.jsx
EVT-04	Event registration flow	Integration	src/features/event/services/
EVT-05	Event categories & filters	UI/Functional	src/features/event/
EVT-06	Event chatbot (jika ada)	UI/Functional	src/layouts/event/chatbot/

F. Volunteer Management (MEDIUM PRIORITY)

ID	Item	Tipe	Modul Path
VOL-01	Public volunteer info	UI/Visual	src/components/volunteer/
VOL-02	Volunteer registration form	UI/Functional	src/features/volunteer/
VOL-03	Admin: Volunteer list	UI/Functional	src/page/dashboard/VolunteerPage.jsx
VOL-04	Volunteer approval/rejection	Integration	src/features/volunteer/services/
VOL-05	Volunteer profile matching	Integration	src/features/volunteer/api/

G. User Profile (LOW PRIORITY)

ID	Item	Tipe	Modul Path
PROF-01	View own profile	UI/Functional	<code>src/page/root/ProfilePage.jsx</code>
PROF-02	Edit profile	UI/Functional	<code>src/features/profile/</code>
PROF-03	Profile picture upload (ImageKit)	Integration	<code>react-cropper</code> + <code>imagekitio-react</code>
PROF-04	Profile picture crop	UI/Functional	<code>react-cropper</code>

H. Admin Dashboard (MEDIUM PRIORITY)

ID	Item	Tipe	Modul Path
ADM-01	Admin dashboard home	UI/Functional	<code>src/page/dashboard/DashboardPage.jsx</code>
ADM-02	User management (admin only)	UI/Functional	<code>src/page/dashboard/UserPage.jsx</code>
ADM-03	Campaign management (admin + bismar)	UI/Functional	<code>src/page/dashboard/CampaignPage.jsx</code>
ADM-04	Donation history (admin + bismar)	UI/Functional	<code>src/page/dashboard/DonationHistory.jsx</code>
ADM-05	Campaign categories	UI/Functional	<code>src/page/dashboard/CampaignCategories.jsx</code>
ADM-06	Proposal management	UI/Functional	<code>src/page/dashboard/proposal/</code>
ADM-07	Products (soon)	—	<code>src/page/dashboard/products/</code>
ADM-08	Orders (soon)	—	<code>src/page/dashboard/orders/</code>

2.2. Item yang TIDAK Akan Diuji

- **IFL Backend API** (external) — di luar scope, hanya black-box test
- **Midtrans Payment Gateway** (3rd party) — hanya test integration, bukan internal Midtrans
- **Google OAuth provider** — 3rd party, hanya test callback flow
- **ImageKit CDN** — 3rd party, hanya test upload/transform response

- **Email delivery** (verification, forgot password) — out of scope
- **Mobile native app** — tidak ada, hanya web responsive
- **Cross-browser legacy** (IE11) — target browser modern
- **Performance under 10K concurrent** — akan diukur terpisah
- **Penetration testing** — out of scope

2.3. Definisi Istilah Penting

Istilah	Definisi
IFL	Indonesian Future Leaders — organisasi
Chapter	Cabang daerah (Malang)
Chatbot Donasi	UI conversational untuk input nominal donasi
Midtrans	Payment gateway Indonesia (Snap, Core API)
Snap	Midtrans pop-up payment UI
QRIS	Quick Response Code Indonesia Standard (payment)
Campaign	Program donasi yang punya target nominal
Bismar	Role internal IFL untuk tim campaign/donasi
Copywriter	Role internal IFL untuk content writer
CKEditor	Rich text editor untuk blog admin
ImageKit	CDN + image transformation service
ProtectedRoute	HOC React Router untuk guard role-based
React Query	Library untuk server state management

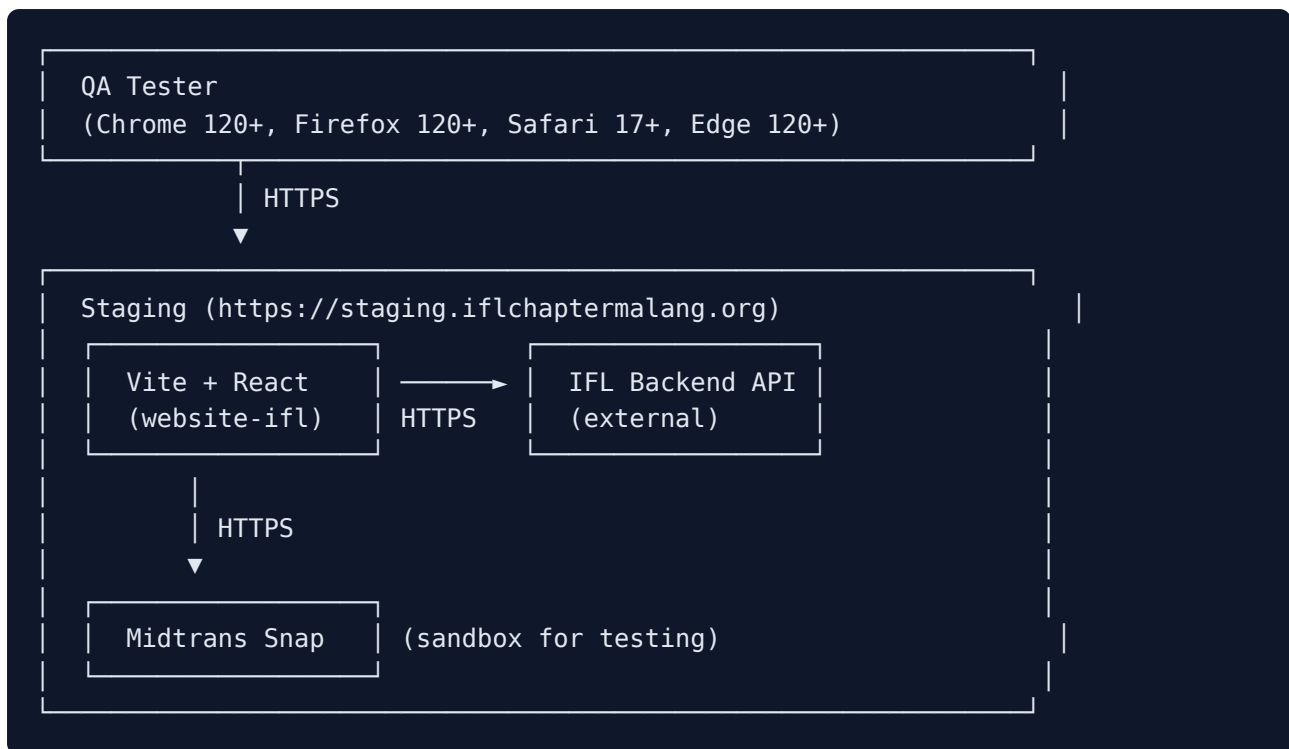
3. Special Terminology (Istilah Khusus)

Istilah	Definisi
Snap Token	Token dari Midtrans untuk membuka pop-up payment
Transaction ID	ID unik untuk satu transaksi donasi
Settlement	Proses Midtrans mengirim dana ke rekening IFL
Donor	User yang berdonasi
Quick reply	Tombol pilihan cepat di chatbot
Composer	Area input user di chatbot
Validation feedback	Pesan error/success di chatbot
CDA	Continuous Donor Acquisition
Recurring donation	Donasi rutin (bulanan) — planned feature
Voucher	Kode diskon / promo donasi
Token JWT	JSON Web Token untuk authentication (custom impl)
SPA	Single Page Application — semua routing client-side

4. Testing Location (Lokasi Pengujian)

Environment	URL	Tujuan	Akses
Local dev	http://localhost:5173 (Vite default)	Development	Developer laptop
Staging	(to be set up)	Pre-production QA	QA team
Production	https://iflchaptermalang.org/	Smoke test only	Public
Midtrans sandbox	https://app.sandbox.midtrans.com	Payment testing	Test cards
Mock backend	MSW (Mock Service Worker)	Unit test isolation	Test runner

Diagram setup pengujian:



5. Testing Schedule (Jadwal Pengujian)

No	Kegiatan	Tanggal Mulai	Tanggal Selesai	PIC	Status
1	Test Plan Review	2026-06-22	2026-06-22	Irza	✅ Done
2	Phase 0: Setup test framework (Vitest + RTL + Playwright)	2026-06-23	2026-06-25	Dev	🕒 Planned
3	Phase 1: Smoke test critical paths (E2E)	2026-06-26	2026-06-28	QA	🕒 Planned
4	Phase 2: Auth & role-based	2026-06-29	2026-07-01	QA	🕒 Planned
5	Phase 3: Donasi + Midtrans	2026-07-02	2026-07-08	QA	🕒 Planned
6	Phase 4: Blog + CKEditor	2026-07-09	2026-07-13	QA	🕒 Planned
7	Phase 5: Event + Volunteer	2026-07-14	2026-07-18	QA	🕒 Planned
8	Phase 6: Public pages	2026-07-19	2026-07-22	QA	🕒 Planned
9	Phase 7: Admin dashboard	2026-07-23	2026-07-27	QA	🕒 Planned
10	Regression Test (full)	2026-07-28	2026-07-30	QA	🕒 Planned
11	UAT (User Acceptance Test)	2026-07-31	2026-08-01	Stakeholder	🕒 Planned
12	Sign-off & Release v2.0	2026-08-02	2026-08-02	PM	🕒 Planned

6. Testing Quality Criteria (Kriteria Kualitas Pengujian)

Kriteria	Target
Kesiapan environment (staging)	100% deployed & smoke test passed
Ketersediaan test data	Akun 3 role tersedia di staging
Build stability	Build sukses tanpa critical blocker
Kesiapan test framework	0% → 100% dalam Phase 0
Code coverage (unit)	0% → 30% di akhir Phase 1, 50% di akhir
E2E coverage	Minimal 1 happy-path + 1 sad-path per fitur critical
Dokumentasi	Test plan + test summary report

7. Entry Criteria (Kriteria Masuk)

- [x] Test plan sudah dibuat dan di-review
- [] **Test framework sudah terinstall** (Vitest + RTL + Playwright) — Phase 0
- [] **Test config files** sudah ada (`vitest.config.js` , `playwright.config.js`) — Phase 0
- [] **CI/CD untuk test** sudah ada di `.github/workflows/` — Phase 0
- [] **Test data** (3 akun role: admin, bismar, copywriter) tersedia di staging
- [] **Midtrans sandbox** account aktif dengan test cards
- [] Tools testing terinstall (Vitest, Playwright, Postman, VS Code)
- [] Backend IFL running di staging
- [] ImageKit credentials tersedia
- [] Google OAuth client ID/Secret tersedia

8. Exit / Suspension Criteria (Kriteria Keluar / Penangguhan)

Suspension (pengujian ditunda): - Build gagal deploy ke staging - Backend IFL API down > 30 menit
 - Midtrans sandbox down - Ada critical bug yang block > 50% test case Donasi - ImageKit CDN error

Resume (pengujian dilanjutkan): - Build sudah stabil - Backend & payment gateway sudah normal -
 Critical bug sudah diperbaiki - Environment sudah stabil minimal 1 jam

9. Completion Criteria (Kriteria Penyelesaian)

- [] **Test framework setup 100%** (Vitest + RTL + Playwright + CI)

- ☐ **100% Donasi test case** dieksekusi (highest priority, 13 TC)
- ☐ **100% Auth test case** dieksekusi (10 TC)
- ☐ Minimal **80% test case** total dieksekusi (seluruh fase)
- ☐ **Code coverage $\geq 30\%$** (dari baseline 0%)
- ☐ **0 critical bug** open
- ☐ **0 high-priority bug** open
- ☐ Semua **regression test** passed
- ☐ Test summary report sudah dibuat
- ☐ Sign-off dari stakeholder

10. Environmental Requirements (Kebutuhan Lingkungan)

10.1. Perangkat Keras (Hardware)

Device	Spec	Jumlah
Laptop Tester	RAM 16GB, SSD 512GB, i5 ke atas	1 (QA)
Smartphone (Android)	Android 13+, 6GB RAM (responsive test)	1
Tablet (iPad)	iPad Air, iOS 17+ (responsive test)	1

10.2. Perangkat Lunak (Software)

Software	Versi	Kegunaan
Node.js	18+	Runtime
Google Chrome	120+	Browser testing (primary)
Firefox	120+	Cross-browser
Safari	17+	macOS testing
Edge	120+	Cross-browser
Playwright	1.40+	E2E test runner (planned)
Vitest	1.0+	Unit test runner (planned)
React Testing Library	14+	Component testing (planned)
Postman	Latest	API testing (manual)
Newman	Latest	Postman CLI (automation)
VS Code	Latest	Code editor

10.3. Jaringan

- Internet min. 10 Mbps
- Akses ke Midtrans sandbox & ImageKit CDN

10.4. Konfigurasi Environment

Variable	Value	Keterangan
<code>VITE_API_URL</code>	<code>https://api.iflchaptermalang.org</code>	Backend API
<code>VITE_MIDTRANS_CLIENT_KEY</code>	(sandbox key)	Midtrans Snap
<code>VITE_MIDTRANS_MERCHANT_ID</code>	(sandbox ID)	Midtrans
<code>VITE_IMAGEKIT_PUBLIC_KEY</code>	(public key)	ImageKit upload
<code>VITE_GOOGLE_CLIENT_ID</code>	(OAuth ID)	Google login
<code>VITE_BASE_URL</code>	<code>https://iflchaptermalang.org</code>	Site URL

11. Test Systems (Sistem Pengujian)

Komponen	Deskripsi	Status
Test plan	Dokumen ini (<code>test-plan.md</code>)	✅ Ada
Test framework	Vitest + RTL + Playwright	❌ Belum di-setup
Test config	<code>vitest.config.js</code> + <code>playwright.config.js</code>	❌ Belum ada
Test case UI	(to be created) — akan disimpan di Google Sheet	🕒 Planned
Test case API	(to be created) — akan disimpan di Google Sheet	🕒 Planned
CI/CD pipeline	<code>.github/workflows/test.yml</code>	❌ Belum ada (cuma <code>deploy.yml</code>)
Mock backend	MSW setup	🕒 Planned
Test data	3 akun role (admin, bismar, copywriter) + donor	🕒 Planned
Pre-commit hook	Husky	❌ Belum ada

12. External Factors (Faktor Eksternal)

- **IFL Backend API** — harus running untuk E2E test

- Disimpan terpisah (bukan bagian dari frontend repo)
- Contract stability tergantung tim backend
- **Midtrans Payment Gateway** — 3rd party
- Sandbox tersedia untuk testing (app.sandbox.midtrans.com)
- Reliability: 99.9% SLA (production), bisa ada downtime
- Test cards: tersedia untuk Visa, MC, QRIS, GoPay, OVO, DANA, ShopeePay
- **ImageKit CDN** — image hosting
- Reliability: 99.9% SLA
- Free tier: 20GB bandwidth/bulan
- **Google OAuth** — login sosial
- Reliability: 99.9% SLA
- **cPanel** — hosting (bukan Vercel)
- Deployment via FTP dari GitHub Actions
- Custom domain: iflchaptermalang.org

13. Team Composition (Komposisi Tim)

Nama	Peran	Tanggung Jawab
Irza (Muhammad Irza Dzulhika)	QA Lead / Tester	Test planning, execution, automation
Dev Frontend	Frontend Developer	Bug fixing, test framework setup
Bismar Team	Internal Role	Campaign management (uji alur donasi)
Copywriter Team	Internal Role	Blog content (uji alur blog admin)
Admin IFL	Administrator	UAT, sign-off

External Participants: Member IFL (uji alur volunteer registration)

Hand-off Points: - Developer → QA: setiap sprint / iterasi - QA → Bismar/Copywriter: untuk UAT per role

14. Revision Handover Management (Manajemen Penyerahan Revisi)

- **Version control:** Git (GitHub)
- **Build deployment:** otomatis via cPanel FTP dari [.github/workflows/deploy.yml](https://github.com/workflows/deploy.yml)
- **QA testing window:** minimal 1x24 jam setelah build baru di staging
- **Changelog format:** `` [v1.0.X] 2026-06-22
- **FIXED:** Bug #123 - Donation timeout error

- ADDED: Recurring donation feature
- CHANGED: Redesign chatbot composer
- TESTED: 13 donation test cases executed ``
- **Notifikasi:** GitHub Issues + WhatsApp group IFL Malang

15. Test Suite Execution (Eksekusi Test Suite)

Strategi per fase:

Fase	Test Suite	Strategi	Tools
Smoke	5 critical-path tests	Automated on every PR	Playwright
Donasi regression	13 donasi test cases	Automated on every PR	Playwright + Vitest
Auth regression	10 auth test cases	Automated	Playwright
Functional	Per fitur (per modul)	Manual + Postman untuk API	Manual + Postman
Regression	Full suite	Automated nightly	Playwright + Vitest
Payment	Midtrans sandbox scenarios	Manual + automated	Playwright + Midtrans
Security	OWASP Top 10 checklist	Manual + automated	OWASP ZAP (planned)
Accessibility	axe-core scan	Automated	axe-core (planned)

Cara eksekusi (IFL-specific):

```
# Setup lokal
cd fe_iflchaptermalang
cp .env.example .env.local
# Update VITE_API_URL, VITE_MIDTRANS_CLIENT_KEY, dll di .env.local

# Install dependencies
npm install

# === PHASE 0: Setup test framework (one-time) ===
npm install -D vitest @testing-library/react @testing-library/jest-dom \
  @testing-library/user-event @playwright/test jsdom msw

# Buat vitest.config.js & playwright.config.js (template)

# === Unit tests (setelah setup) ===
npm test                # Run once
npm run test:watch       # Watch mode
npm run test:coverage    # With coverage

# === E2E tests (setelah setup) ===
npx playwright install  # First time only

npm run test:e2e        # All E2E
npm run test:e2e:auth   # Auth flow
npm run test:e2e:donation # Donasi + Midtrans
npm run test:e2e:blog    # Blog + CKEditor
npm run test:e2e:event   # Event management

# === E2E debugging ===
npm run test:e2e:ui      # Interactive UI mode
npm run test:e2e:debug   # Debug mode
npm run test:e2e:headed  # See browser

# === API testing (after Postman collection ready) ===
newman run ifl-api.postman_collection.json
```

16. Potential Risks (Risiko Potensial)

Risiko	Dampak	Probabilitas	Mitigasi
Setup test framework dari 0%	Timeline Phase 0 molor	High	Pakai template yang sudah ada (Arteri E-Learning)
Backend IFL API down	E2E test gagal	Medium	Mock backend dengan MSW untuk unit test
Midtrans sandbox error	Payment test gagal	Low	Backup test cards, retry dengan exponential backoff
JavaScript (no TS)	Bug type tidak tertangkap saat compile	High	Strict ESLint + PropTypes + gradual TS migration
ImageKit CDN rate limit	Image upload test gagal	Low	Mock ImageKit response di test
Tight volunteer	Coverage tidak maksimal	High	Risk-based testing; fokus Donasi (highest impact)
Tim QA single person (Irza)	Bottleneck di review	High	Pair dengan dev untuk code review test
CKEditor 5 bug	Blog authoring broken	Low	Pin versi, test scenario terbatas
3 icon library (Phosphor + Lucide + React Icons)	Bundle bengkak, conflict	Medium	Audit & konsolidasi ke 1 library
cPanel FTP deploy	Build lama, error-prone	Medium	Monitor deploy log, retry script
Google OAuth client quota	Login test gagal	Low	Mock OAuth di unit test
No TypeScript	Runtime error lebih sering	High	Strict prop validation + JSDoc

17. Documentation of Changes (Dokumentasi Perubahan)

Versi	Tanggal	Penulis	Perubahan
v1.0	2026-06-22	Irza	Initial Test Plan — berdasarkan analisis codebase IFL Chapter Malang

18. Appendix (Lampiran)

Dokumen pendukung: - **Source repo** — https://github.com/irzadzulhika29/fe_iflchaptermalang - **Live site** — <https://iflchaptermalang.org/> - **Test plan template** — [../../docs/test-plan-template.md](#) - **Studi reference** — [../arteri-elearning/test-plan.md](#) (pembanding) - **JS testing patterns** — [../../skills/javascript-testing-patterns/SKILL.md](#) - **E2E patterns** — [../../skills/e2e-testing-patterns/SKILL.md](#)

19. FAQ (Pertanyaan Umum)

Q: Kenapa test plan ini ditulis padahal test belum ada? A: Karena **best practice** — test plan harus dibuat sebelum coding test. Tanpa test plan, setup test framework bisa salah arah. Dokumen ini jadi panduan Phase 0.

Q: Kenapa prioritas Donasi (bukan Auth)? A: Karena **Donasi = revenue stream + payment gateway integration**. Bug di donasi = kerugian finansial langsung. Auth juga penting, tapi biasanya lebih mature pattern-nya.

Q: Bagaimana cara test Midtrans di lokal? A: Midtrans punya **sandbox** terpisah ([app.sandbox.midtrans.com](#)). Pakai client key sandbox + test cards (Visa: 4811 1111 1111 1114, dll). Transaksi sandbox tidak benar-benar dipotong.

Q: Kenapa tidak ada test sama sekali? Apa kendalanya? A: Belum sempat di-setup. Project fokus ke feature delivery dulu. Sekarang dengan test plan ini, mulai di-setup fondasi (Phase 0).

Q: JavaScript atau TypeScript — perlu migrasi? A: Untuk **jangka panjang** TypeScript sangat membantu (type safety, IntelliSense, fewer bugs). Untuk **jangka pendek** (magang QA), cukup strict ESLint + PropTypes. Migrasi TS bisa jadi proyek terpisah.

Q: 3 icon library (Phosphor + Lucide + React Icons) — kenapa? A: Kemungkinan karena refactor bertahap. Bundle jadi bengkak. Quick win: konsolidasi ke 1 library (rekomendasi: **Lucide** — paling ringan & modern).

Q: Berapa lama test plan ini berlaku? A: Berlaku untuk **v2.0** (Phase 0-12 ≈ 6 minggu). Update setiap ada perubahan arsitektur besar atau setelah milestone tercapai.

Q: Apakah ada CI/CD untuk test? A: **Belum**. Saat ini cuma ada [deploy.yml](#) (FTP deploy). Akan ditambah [test.yml](#) di Phase 0.

Referensi

- Mulyanto, A. (2018). *Test Plan*. BINUS University – School of Information Systems.
- Myers, G. J., Sandler, C., & Badgett, T. (2012). *The Art of Software Testing* (3rd ed.).
- IEEE 829 — Standard for Software Test Documentation
- ISTQB Foundation Level Syllabus v4.0
- React Testing Library Docs — <https://testing-library.com/docs/react-testing-library/intro>

- Playwright Documentation — <https://playwright.dev/docs/intro>
- Vitest Documentation — <https://vitest.dev/>
- Midtrans Sandbox — <https://docs.midtrans.com/en/technical-reference/sandbox-test>

Change Log (Dokumen Ini)

Versi	Tanggal	Perubahan
v1.0	2026-06-22	Initial Test Plan — di-instantiate dari template BINUS, diisi spesifik untuk IFL Chapter Malang. Fokus utama: setup test framework dari 0% + donasi flow + role-based access.